Natalie Maciel
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Education

California State University – Fresno
Masters of Science - Mechanical Engineering — Graduation: May 2027
Bachelors of Science - Mechanical Engineering — Graduation: May 2026
Honors: President's List: Spring 2024 | Dean's List: Fall 2024, Spring 2025, Fall 2025, Spring 2026

Professional Employment

Systems Design Intern
Galen Ko, Northrop Grumman, Northridge, CA — *June 2026 - Present*
- Proposed subsystem requirements to be rewritten, removed, or restructured for missile defense systems to improve the efficiency of development and reduce the overall cost for the program
- Created procedural documents to help integrate the entire project architecture for SiAW and delegate systems
- Utilized DOORS to connect, organize, and structure requirements for SiAW as specifications continued to change
- Authored change notices for documentation in Teamcenter, updating revisions and assigning specialists to sign off
- Automated requirements data in DOORS for monthly data collection and analytical tracking

AY Teaching Associate Engineering Graphics
Fresno State, Lyles College of Engineering, Fresno, CA — *August 2026 - Present*
- Supervised and guided up to 33 first-year engineering students through in-class lab challenges and semester projects.
- Lectured on strong CAD practices including GD&T, 2D and 3D SolidWorks, motion studies, and 2D and 3D AutoCAD

Manufacturing Intern
Saman Ibrahim, Northrop Grumman, Northridge, CA — *May 2025 - July 2025*
- Optimized manufacturing processes within the assembly line, improving manufacturing efficiency for AARGM
- Initiated purchase orders and worked with Supply Chain to obtain hardware within a reasonable timeframe at a pre-approved cost
- Addressed nonconformances and proposed rework practices to improve production and quality of defense systems
- Collaborated with quality engineers, program management, and technicians to better organize and address common bottlenecks within existing manufacturing processes

Engineering Design Intern
Ryan Carlson, Lawrence Engineering Group, Fresno, CA — *April 2024 - April 2025*
- Modeled 40+ architectural designs in Hourly Analysis Program and EnergyPro to test Title 24 energy standards
- Designed ductwork placement for 10+ new non-residential buildings across the Central Valley using Bluebeam

STEM Tutor
Self-Employed, Clovis, CA — *March 2020 - May 2025*
- Designed custom lesson plans and practice exams focused on individual student needs
- Implemented structured progress tracking and evolving lesson plans that adapted to students' needs to ensure progress

Projects

Biomedical Research - Reducing Noise Pollution in Endoscopy Suite — *August 2023 - January 2024*
- Created SolidWorks models of different springs and spring covers to dampen noise pollution in endoscopy units
- Researched internal mechanisms of endoscopes to pinpoint locations of significant noise in the design

SunDogs Club Project - Solar Powered Charging Station — *January 2024 - August 2025*
- Modeled a structural design for solar panels to have a point of rotation so the panels had constant sun exposure
- Collected data from solar panels to analyze the amount of generated power and the number of electric scooters being supported on a single charge

AVI Robotics Club Project - Propulsion and Chassis Design Team — *August 2025 - Present*
- Prototyped SolidWorks parts for the tether, strain relief, and legs of an ROV system
- Performed a buoyancy calculation on the tether and strain-relief system to ensure neutral buoyancy in the system
- Initiated material procurement processes to ensure designs were feasible with existing financial and time restrictions

UGV Research - Peach Harvesting Device — *August 2025- Present*
- Used embedded control systems to communicate with subsystems being controlled through C++
- Designed an end effector in SolidWorks, conducting motion studies to observe performance
- Organized software components using GitHub

Technical Skills & Coursework

- **Technical Skills:** MATLAB, C++, SolidWorks, AutoCAD, EnergyPro, Bluebeam, Hourly Analysis Program, Arduino, DOORS, Siemens Teamcenter